

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: MLA03

COURSE CODE: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO2: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Dr G.A. SENTHIL

QUESTIONS: 5

Total Marks: 50

Annexure A
Application - Based Questions

Duration: 2Hrs

Code of Conduct:

I ALLAPAREDDY JAHNAVI (192472214) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Jahnavi

1) Reinforcement Learning Framework for a Ride-Sharing System

1. Objective

The objective of the ride-sharing system is to efficiently match available drivers with passenger requests while minimizing passenger waiting time, reducing travel distance, and improving overall service quality. The system aims to maximize driver utilization and enhance customer satisfaction through intelligent decision-making.

2. Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning is an appropriate approach because the ride-sharing environment is dynamic, with continuously changing driver locations, passenger requests, and traffic conditions. RL enables the system to learn the best matching strategy through continuous interaction with the environment and improve decision-making based on rewards received from previous actions.

3. Environment and Agent

Environment

- Driver locations
- Passenger requests
- Road network
- Traffic conditions
- Driver availability

Agent

- Ride-sharing matching system that selects the most suitable driver for each passenger request.

4. State Space

The state space represents the current status of the ride-sharing system.

It includes:

- Current driver locations
- Passenger pickup locations
- Passenger destinations
- Traffic conditions
- Driver availability
- Estimated travel time
- Number of active ride requests

5. Action Space

The available actions for the RL agent include:

- Assign the nearest available driver
- Assign an alternative nearby driver
- Reassign a driver
- Reject a ride request if no drivers are available
- Delay assignment until a better driver becomes available

6. Reward Function

Positive Rewards

- Reduced passenger waiting time
- Shorter driver travel distance
- Successful ride completion
- High customer ratings
- Efficient driver utilization

Negative Rewards

- Long passenger waiting time
- Increased travel distance
- Ride cancellation
- Driver idle time
- Poor customer ratings

7. Policy

The policy is the decision-making strategy used by the RL agent to select the best driver for each passenger request. Based on the current state of the system, the agent chooses the action that maximizes long-term rewards by considering factors such as driver proximity, traffic conditions, and availability.

8. Learning Process

The RL agent learns by interacting with the ride-sharing environment. After assigning a driver, it receives rewards based on the quality of the decision, such as reduced waiting time or successful ride completion. The agent continuously updates its policy using these rewards, enabling it to make more accurate driver assignments over time.

9. RL Architecture / Block Diagram



10. Expected Outcomes

- Reduced passenger waiting time
- Efficient driver utilization
- Faster ride allocation
- Improved service quality
- Increased customer satisfaction
- Better traffic-aware decision-making
- Higher operational efficiency
- Continuous improvement through learning

2) Reinforcement Learning Framework for a Robotic Vacuum Cleaner

1. Objective

The objective is to enable a robotic vacuum cleaner to clean an environment efficiently while avoiding obstacles, minimizing cleaning time, reducing energy consumption, and ensuring complete floor coverage through intelligent decision-making.

2. Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning is suitable because the robot operates in a dynamic environment where furniture placement, obstacles, and room layouts may change. RL enables the robot to learn efficient cleaning strategies through trial and error without requiring manually programmed paths.

3. Environment and Agent

Environment

- Rooms
- Furniture
- Obstacles
- Floor layout
- Charging station

Agent

- Robotic vacuum cleaner that decides its movement and cleaning actions.

4. State Space

The state space includes:

- Current robot location
- Battery level
- Obstacle positions
- Dirty and clean areas

- Distance to charging station

5. Action Space

The available actions include:

- Move forward
- Turn left
- Turn right
- Start cleaning
- Return to charging station
- Stop cleaning

6. Reward Function

Positive Rewards

- Successfully cleaned an area
- Avoided obstacles
- Reduced cleaning time
- Efficient battery usage
- Returned safely to charging station

Negative Rewards

- Collision with obstacles
- Repeated cleaning of the same area
- Battery depletion
- Missing dirty areas
- Long cleaning time

7. Policy

The policy determines the best movement and cleaning action based on the robot's current location, battery level, and nearby obstacles to maximize cleaning efficiency.

8. Learning Process

The robot explores different cleaning paths and receives rewards after each action. Over time, it learns the most efficient routes, avoids unnecessary movements, and improves cleaning performance using the learned value function.

9. RL Architecture / Block Diagram



10. Expected Outcomes

- Efficient cleaning
- Complete floor coverage
- Reduced cleaning time
- Lower battery consumption
- Improved obstacle avoidance
- Better autonomous navigation

3) Reinforcement Learning Framework for a Smart Grid System

1. Objective

The objective is to efficiently manage electricity distribution by balancing power generation, storage, and consumption while minimizing energy wastage, reducing operational costs, and ensuring a stable electricity supply.

2. Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning is appropriate because electricity demand and supply change continuously. RL enables the smart grid to learn optimal power distribution strategies by interacting with the environment and maximizing long-term operational efficiency.

3. Environment and Agent

Environment

- Power grid
- Electricity demand
- Renewable energy sources
- Battery storage
- Consumers

Agent

- Smart grid energy management system.

4. State Space

The state space includes:

- Current electricity demand
- Power generation
- Battery storage level
- Renewable energy availability
- Grid load

5. Action Space

The available actions include:

- Increase electricity supply
- Reduce electricity supply
- Store excess energy
- Release stored energy
- Switch between energy sources

6. Reward Function

Positive Rewards

- Stable electricity supply
- Efficient energy utilization
- Reduced energy wastage
- Lower operational costs
- Balanced power distribution

Negative Rewards

- Power outages
- Energy wastage
- Battery overloading
- High operating costs
- Grid instability

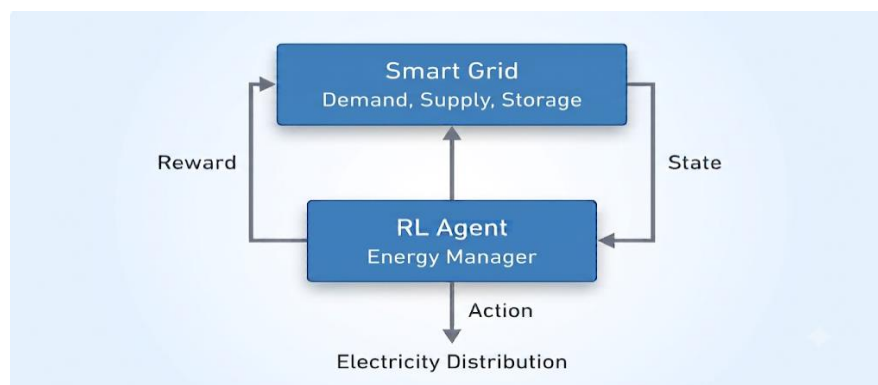
7. Policy

The policy selects the most efficient energy distribution strategy according to electricity demand, renewable energy availability, and battery status.

8. Learning Process

The RL agent continuously observes demand and supply conditions, performs energy management actions, receives rewards, and updates its policy to improve future electricity distribution decisions.

9. RL Architecture / Block Diagram



10. Expected Outcomes

- Stable electricity supply
- Reduced energy wastage
- Improved grid reliability
- Efficient energy utilization
- Lower operational costs
- Better renewable energy integration

4) Reinforcement Learning Framework for a Drone Delivery System

1. Objective

The objective is to enable a drone delivery system to determine the safest and shortest delivery routes while avoiding obstacles, minimizing delivery time, and reducing energy consumption.

2. Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning is suitable because drone navigation involves changing weather conditions, moving obstacles, and varying delivery locations. RL allows the drone to continuously learn efficient navigation strategies through interaction with the environment.

3. Environment and Agent

Environment

- Delivery locations
- Buildings
- Trees
- Weather conditions
- Airspace

Agent

- Autonomous delivery drone.

4. State Space

The state space includes:

- Drone location
- Destination
- Battery level
- Wind speed
- Nearby obstacles
- Weather conditions

5. Action Space

The available actions include:

- Fly forward
- Turn left
- Turn right
- Change altitude
- Hover
- Return to base

6. Reward Function

Positive Rewards

- Successful delivery
- Short delivery time
- Safe navigation
- Efficient battery usage
- Obstacle avoidance

Negative Rewards

- Collision
- Delivery delay
- Battery depletion
- Wrong route
- Failed delivery

7. Policy

The policy enables the drone to choose the safest and most efficient flight path based on the current environment and delivery destination.

8. Learning Process

The drone explores different routes during navigation and learns from successful and unsuccessful deliveries. Through exploration and exploitation, it continuously improves route selection and delivery efficiency.

9. RL Architecture / Block Diagram



10. Expected Outcomes

- Faster deliveries
- Optimized route planning
- Reduced battery consumption
- Safe obstacle avoidance
- Improved delivery success rate
- Higher customer satisfaction

5) Reinforcement Learning Framework for a Smart Irrigation System

1. Objective

The objective is to optimize water usage by automatically controlling irrigation based on soil moisture, weather conditions, and crop requirements while minimizing water wastage and improving agricultural productivity.

2. Why Reinforcement Learning is an Appropriate Approach

Reinforcement Learning is suitable because environmental conditions such as rainfall, temperature, and soil moisture change continuously. RL enables the irrigation system to learn the optimal watering schedule through continuous interaction with the environment.

3. Environment and Agent

Environment

- Agricultural field
- Soil moisture
- Weather conditions
- Crops
- Water supply

Agent

- Smart irrigation controller.

4. State Space

The state space includes:

- Soil moisture level
- Temperature
- Humidity
- Rainfall prediction
- Crop growth stage
- Water availability

5. Action Space

The available actions include:

- Start irrigation
- Stop irrigation
- Increase water flow
- Reduce water flow
- Delay irrigation

6. Reward Function

Positive Rewards

- Optimal soil moisture
- Water conservation
- Healthy crop growth
- Reduced irrigation cost
- Increased crop yield

Negative Rewards

- Water wastage
- Over-irrigation
- Under-irrigation
- Crop stress
- Increased operational cost

7. Policy

The policy selects the best irrigation action based on current soil moisture, weather conditions, and crop requirements to maximize water efficiency and crop productivity.

8. Learning Process

The RL agent continuously monitors field conditions, performs irrigation actions, receives rewards based on crop health and water usage, and updates its policy to improve future irrigation decisions.

9. RL Architecture / Block Diagram



10. Expected Outcomes

- Efficient water utilization
- Improved crop yield
- Reduced water wastage
- Lower irrigation costs
- Sustainable agricultural practices
- Better decision-making through continuous learning

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand